

Methods for Observing Human Experience

- Innovation starts and ends with people.
- The process of Human-Centered Design involves investigating people in their natural setting.
- This requires curiosity, objectivity, and empathy.



ETHNOGRAPHIC RESEARCH

- **Stepping out of your native environment to learn what people do in the places they inhabit** is a great way to discover opportunities for innovation.
- The methods in this grouping are good for studying human behavior in its natural setting.
- Interviewing
- Fly-on-the-Wall Observation
- Contextual Inquiry
- Walk-a-Mile Immersion

Interviewing

A technique for gathering information through direct dialogue.

- A good interview helps you take advantage of this natural inclination in order to gather valuable information.
- Interviewing gives you an **opportunity to speak directly with the people who can help you make informed decisions.**
- Through these interviews you **gain a better sense of people and their views of the world** by subtly eliciting their true feelings, desires, struggles, and opinions through a few carefully crafted questions.
- **A good interviewer needs to be attuned to the interviewee** to know when to probe for more information, when to redirect the conversation, and how to parse what is meant from what is said.

Quick Guide

- Identify a topic for investigation.
- Prepare your questions and recording equipment.
- Determine your criteria for selecting interviewees.
- Identify the people you will interview.
- Set a time and place to meet them.
- Introduce yourself and the purpose. Obtain consent.
- Start with easy questions, then draw out specifics.
- Listen carefully and take good notes.
- Thank each participant.



Helpful Hints:

- Try to choose a location with minimal distractions.
- Don't put words into the interviewee's mouth.
- Resist the urge to conduct an analysis at this stage.

Benefits:

- Helps you gain information directly
- Challenges your preconceptions
- Deepens your empathy for others
- Builds credibility with stakeholders

Fly-on-the-Wall Observation

An approach to conducting field research in an unobtrusive manner

- In situations where you cannot speak directly with people, or do not want to interrupt the flow of their activities, being a fly on the wall has its advantages.
- You'll come to find that careful, unobtrusive observation provides valuable insight you cannot otherwise obtain.
- When left to their own devices, people are likely to say or do things that they're not aware of and would not be able to articulate, even if prompted.
- If you can watch and listen without interfering, you have a chance to capture people's natural behavior.

Quick Guide:

- Identify a subject area to study.
- Develop a plan to guide your investigation.
- Consider which people and activities to watch.
- Choose a location to visit.
- Obtain the necessary access and permission(s).
- Prepare materials for capturing what you see.
- Go out and observe.
- Record your findings in videos, photos, and notes.



- Remember to pay careful attention to people's tasks and workflow, taking note of the information, tools, and people they rely upon to do what they do.
- Also be mindful of the surrounding environment, understanding that peripheral objects, sounds, and people may affect outcomes.

Helpful Hints:

- Make every effort to blend in to the background.
- Take on the role of an objective bystander.
- Look at the situation from several vantage points.

Benefits:

- Diminishes your presence as a researcher
- Deepens your empathy for others
- Challenges your assumptions
- Informs subsequent research activities

Walk-a-Mile Immersion

A way of building empathy for people through firsthand experience

- AS ATTICUS FINCH teaches us in *To Kill a Mockingbird*, “You never really understand a person until you consider things from his point of view—until you climb into his skin and walk around in it.”
- While any research method can help you better understand people and their needs, Walk-a-Mile forces you to take a person’s journey and experience their joys, conflicts, and weariness.
- In other words, you must not only see, but also feel what it is like to live in the world as someone else.

Quick Guide:

- Identify whose experience you want to replicate.
- Choose the tasks and activities you will perform.
- Assemble what is needed to run a simulation.
- Determine the best location.
- Obtain the necessary access and permission(s).
- Conduct the targeted tasks.
- Do each activity as realistically as possible.
- Note your findings along the way.

Helpful Hints

- Commit to the activity fully. Don't give up early.
- Ask another observer to help you capture findings.
- Use an Empathy Suit to simulate human conditions.

Benefits

- Helps you gain firsthand knowledge
- Fosters an attitude of humility
- Deepens your empathy for others
- Informs subsequent research activities

Contextual Inquiry

An approach to interviewing and observing people in their own environment

- “WHAT PEOPLE SAY, what people do, and what they say they do are entirely different things,” observed influential anthropologist Margaret Mead.
- Following this wisdom, it is crucial that we pay attention to what people say and what people do in order to get a clear picture of what really happens.
- A Contextual Inquiry places you in the midst of a person’s environment where you can inquire about his or her experiences in context as they are happening.
- Consequently, input comes directly from the people who have the most knowledge, saving you from making assumptions about how and why things are done.

Quick Guide:

- Identify a location and the people to be involved.
- Prepare your questions and recording equipment.
- Go to the site.
- Introduce yourself and the purpose. Obtain consent.
- Ask the participants to do tasks in a normal way.
- Observe their actions in an unobtrusive manner.
- Interject questions at opportune moments.
- Record your findings in videos, photos, and notes.
- Thank each participant.

Helpful Hints:

- Ask people to do activities, not just give you a tour.
- Use more than one researcher to get multiple views.
- Stay focused on your goals, yet open to discovery.

Benefits:

- Reveals what people actually do and say
- Deepens your empathy for others
- Challenges your assumptions
- Builds credibility with stakeholders

PARTICIPATORY RESEARCH

- Innovators offer solutions that people don't even know they want.
 - This cluster of methods allows you to **engage with your intended audience by equipping them with creative ways to express themselves.**
 - If you pay close attention you'll discover the audience critical and latent needs.
- What's on Your Radar?
 - Buy a Feature
 - Build Your Own
 - Journaling

What's on Your Radar?

An exercise in which people plot items according to personal significance.

- This method **provides a template where people can organize items within a given scope based on how important or relevant they consider them to be.**
- So, whether you are dealing with concrete items or abstract concepts, **the diagram is a useful way for people to assign rank.**
- Reading the diagram is straightforward: **the things people assign to the center circle (which is deliberately small) are most significant, while those in successive circles are less so, and those outside the diagram are not even blips on their radar.**
- The format forces participants to **express clear distinctions between what is primary, what is secondary, and what is tertiary.**

Quick Guide:

- Identify a topic for consideration.
- Make a large poster that looks like a radar screen.
- Include **3 concentric circles and 4-6 segments**.
- **Label the circles: Primary, Secondary, Tertiary.**
- Label the segments as subcategories of the topic.
- Invite a **group of stakeholders to be participants**.
- **Give each person a poster, a pen, and sticky notes.**
- Instruct them to **plot their personal considerations**.
- Ask the participants to describe their rankings.



Helpful Hints:

- Limit the time for plotting items to 15 minutes.
- Allow participants to write in some segment labels.
- Listen closely when people describe what they did.

Benefits:

- Reveals what people are thinking.
- Shows how people prioritize.
- Challenges your preconceptions.
- Yields documents that inform work happening as a consequence.

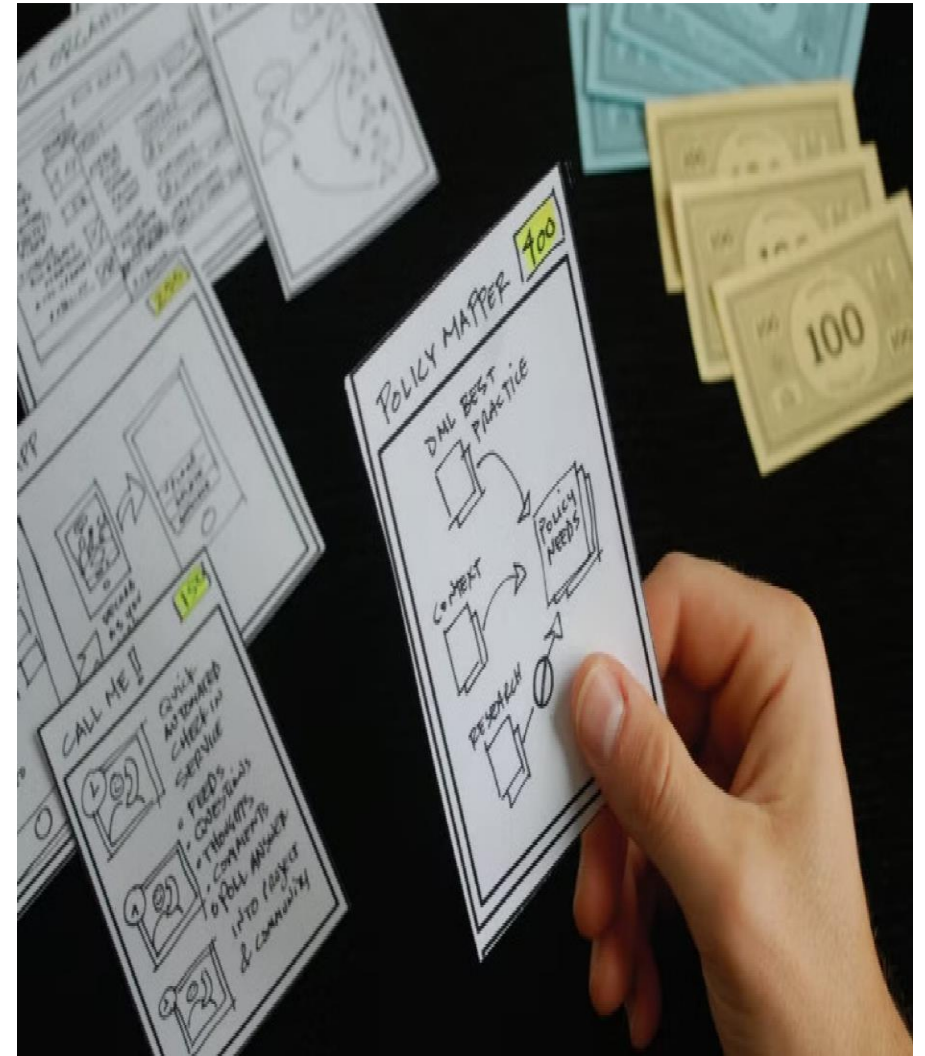
Buy a Feature

A game in which people use artificial money to express trade-off decisions.

- This method is based on a system of constraints that create tension by offering choices that exceed available resources. **It aims to simulate the conditions that exist when people have to budget their resources to get what they truly desire.**
- **Because you provide people with a limited amount of currency with which to buy items, participants must pick and choose which are most important.**
- **The resulting decisions are valuable in assessing what features or concepts should be present in the final design.**
- Additionally, you **can monitor the decision-making process, asking questions about why participants make certain choices.** Those answers may be just as meaningful as their purchases.

Quick Guide:

- Identify a product, service, or policy to focus on.
- Generate a list of potential features.
- **Make playing cards for the various features.**
- **Include a price for each feature.**
- Invite a **group of stakeholders to play the game.**
- **Give each player a set of cards with price tags.**
- **Give them a limited amount of artificial money.**
- **Ask them to purchase features within the budget.**
- Encourage them to clearly express their thoughts, discussions, or considerations.



Helpful Hints:

- Pricing can be based on the actual cost of execution.
- Listen for evidence of motivations and priorities.
- Have participants make buying decisions in pairs.

Benefits:

- Reveals what people value.
- Shows how people deliberate.
- Uncovers latent and unmet needs.
- Yields documents that inform work happening as a consequence.

Build Your Own

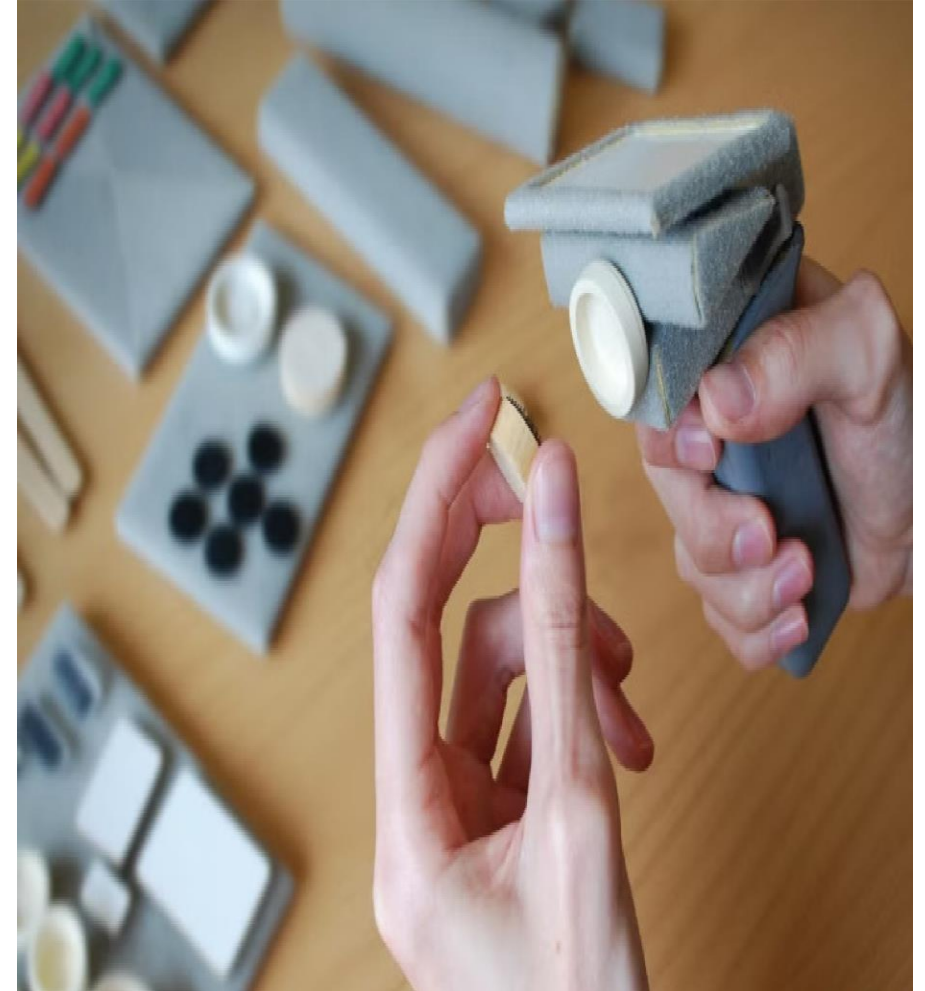
An activity in which people express ideal solutions using symbolic elements.

- **It is a method that helps people express their “what ifs” by putting tools for creativity and communication literally in their hands.**
- **It equips them with a kit of tangible symbolic elements that makes it quick and easy to create a physical representation of an idea.**
- **What you end up with are illuminating artifacts that provide a reference for what they desire or expect of your team’s ultimate solution.**
- **A well-planned, easy to use kit of parts makes it possible for people to build what they imagine.**

- The components you provide need to be simple and relatively nondescript, but should be appropriate to the type of solution you desire.
- For instance, a toolkit comprised of building blocks and figurines could work for urban residents imagine and communicate how they would like to use public spaces such as parks, streets, or market places.
- However, a toolkit consisting of pre-drawn user-interface elements and a sheet of paper would work better for IT professionals imagining their ideal network-monitoring interface.

Quick Guide:

- Identify a product, service, or policy to focus on.
- **Make a kit of representational building blocks.**
- Include a variety of basic shapes and symbols.
- Invite a group of primary stakeholders to participate.
- **Divide the group into teams of two people.**
- **Give each team a construction kit.**
- **Ask them to build an expression of an ideal solution.**
- Encourage them to “think aloud” as they construct.
- **Ask each team to present their final model.**



Helpful Hints:

- Make units easy to build with magnets or Velcro.
- Limit the amount of time for building (15-30 min).
- Listen carefully as teams express wants and needs.

Benefits:

- Shows what people want and desire.
- Uncovers latent and unmet needs.
- Challenges your assumptions.
- Yields models to inform subsequent work.

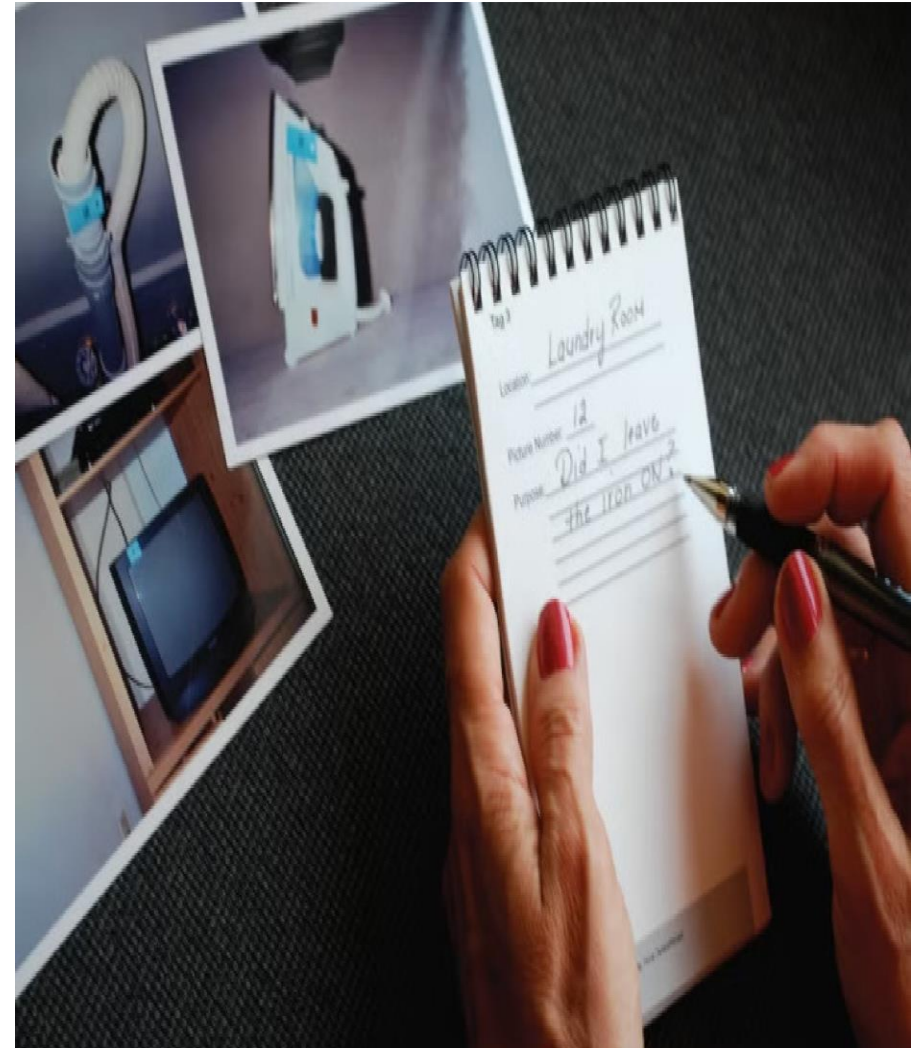
Journaling

An activity that invites people to record personal experiences in words and pictures.

- As a research method, Journaling is a **powerful way to learn about the inner workings of people as they document their personal experiences with a particular product or issue.**
- In contrast to activities that require face-to-face interaction, **journaling is done privately, typically over the course of days or weeks.**
- Often **people will share greater detail about their feelings and opinions** when they do not have to do it in person, yielding very thoughtful and thought-provoking responses.
- A **journaling activity could ask participants to take photographs of their interactions and describe them, narrate a series of short videos, or provide written responses to open-ended prompts.**
- Whatever the chosen tools, craft them carefully to facilitate good findings.

Quick Guide:

- **Identify a subject area to study.**
- Make a kit of materials for record keeping.
- **Include a paper diary** and/or access to a blog.
- **Invite a group of primary stakeholders to participate.**
- **Explain the purpose and duration of the study.**
- **Distribute the kits** with simple instructions.
- **Include a guide for capturing pictures and video.**
- **Ask them to fill the journal and send it back to you.**
- **Perform an exit interview with each participant.**



Helpful Hints:

- Take advantage of the devices people already carry.
- Send periodic reminders to create journal entries.
- Provide the postage needed for returning the kits.

Benefits:

- Accumulates research from remote regions.
- Gains information over time.
- Reveals what people think and feel.
- Deepens your empathy for others.

EVALUATIVE RESEARCH

- When you look at things critically, with an eye toward improvement, you set your course in the direction of making things better.
 - These methods are **good for assessing the usefulness and usability of solutions meant to serve people in new and better ways.**
 - It involves **collecting and analyzing data to determine whether a particular initiative has achieved its intended goals.**
- Think-Aloud Testing
 - Heuristic Review
 - Critique
 - System Usability Scale

Think-Aloud Testing

A testing format where people narrate their experience while performing a given task.

- This style of **evaluation** doesn't just ask for a **play-by-play**, but rather a **thought-by-thought** account of an experience.
- People's articulation of a typically unspoken thought process not only **helps the team pinpoint where the success and failure points are in their system**, but it also **diagnoses the causes**.
- When successful, a Think-Aloud Test **reveals important insights and inferences** that would otherwise be unknown to a design team.

- **Effective sessions**—those in which comfortable reviewers offer lots of clear information—require the sensitive moderation of a well- prepared testing team.
- **By understanding peoples mindset**, intentions, and expectations, which will in turn help you **determine what specific attributes of your design are causing certain reactions and how to address them.**
- Fortunately **just six to nine of these tests will usually reveal 80 % of the issues with a design**, so it is a **low-investment opportunity to make a significant impact on the efficiency, effectiveness, and satisfaction of a solution.**

Quick Guide:

- **Identify what you will be testing** and a few key tasks.
- Invite **6-9 different people to be the respondents**.
- **Schedule a testing session with each person**.
- Introduce yourself and the purpose. Obtain consent.
- Remind each respondent, “We are NOT testing you.”
- Instruct them to conduct each task one at a time.
- Ask them to think aloud.
- Keep quiet, listen carefully, and take good notes.
- Thank each participant.



Helpful Hints:

- Delay any direct questions until the end of the test.
- Avoid the temptation to conduct a demonstration.
- Mimic functionality if the design is still in progress.

Benefits:

- Reveals what people are thinking
- Deepens your empathy for others
- Uncovers opportunities for improvement
- Lowers development costs through early discovery

Heuristic Review

An auditing procedure based on ten rules of thumb for good design.

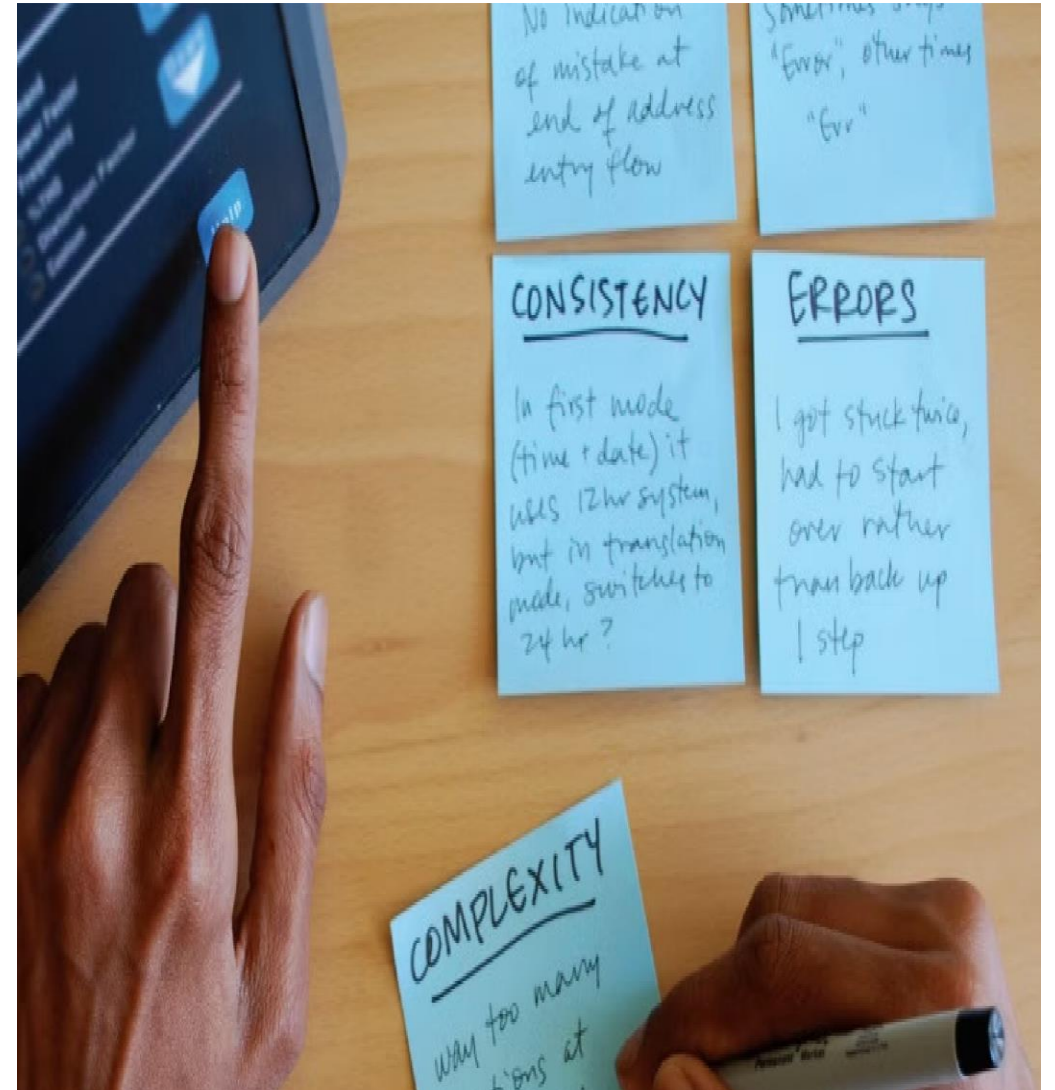
- A heuristic is also known as a rule of thumb—a **generally accurate guideline based on the experiential knowledge of how something works best.**
- We use countless heuristics day-to-day to help simplify our lives and avoid complications. Rules like these provide a quick safety check in our daily activities.
- In much the same way, **designers use heuristics as a way to formulate and evaluate solutions.**

- Drawing from experience, experts have learned that a design's failure or success can often be predicted by how well it addresses the following principles of good design:
 - 1.Match mental model:** A good design should work the way users expect it to work based on their past experiences.
 - 2.Minimize perceived complexity:** Design should look simple.
 - 3. Use consistent form, words, and actions:** The same elements should look and behave the same across the system.
 - 4.Provide a sense of place:** Users should always know where they are in the system and how they reached there.
 - 5.Account for user and environmental constraints:** product works well in real-world situations.
 - 6.Anticipate needs:** Good design predicts what users might want next and provides support in advance.

7. **Use clear and concise language:** Instructions, labels, and messages should be simple and easy to understand.
8. **Give feedback about actions and status:** system should always respond to user actions.
9. **Prevent errors and provide graceful recovery:** design should stop users from making mistakes whenever possible.
10. **Strive for appropriate and minimal aesthetics:** The interface should be visually pleasing.

Quick Guide:

- Identify the subject of your review.
- Form a team of reviewers with multiple perspectives.
- Get everyone familiar with the ten heuristics.
- Select a small number of key tasks.
- Instruct each reviewer to conduct each task.
- Remind them to keep the heuristics in mind.
- Give each reviewer a pen and a sticky note pad.
- Ask them to note all of the issues they discover.
- Tell them to cite one heuristic for each issue.



Helpful Hints:

- Ask the reviewers to add their initials to the notes.
- Encourage them to describe each issue clearly.
- Discourage the inclusion of solutions at this stage.

Benefits

- Leverages proven principles of good design
- Helps you identify problems quickly
- Yields data in the absence of test participants
- Shows opportunities for improvement

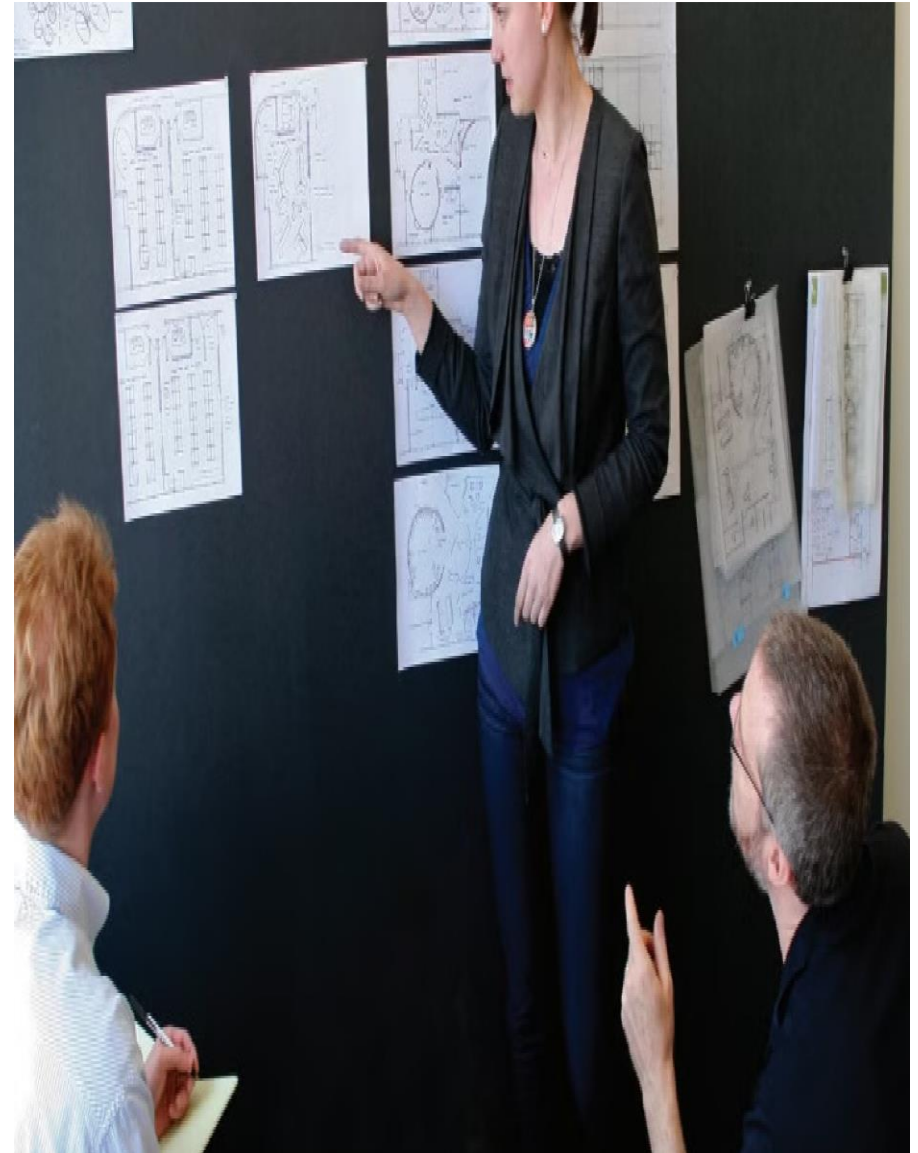
Critique

A forum for people to give and receive constructive feedback

- “Critical feedback is most effective when it is audible, credible, and actionable.”
- Critiques, therefore, should **follow a structure that encourages efficient, productive discussion, thus inspiring collaboration and objectivity and advancing improvements more quickly.**
- As opposed to casual conversation about a project, **a Critique allows the designers to present their current solution and then express concerns or ask specific questions about it.**
- Reviewers are invited to respond in a clear way that addresses the designers’ needs.
- In the context of a **structured Critique— which allows for both positive and negative feedback**—people are more likely to share suggestions for improvement, since the design team has formally solicited them.
- **A good Critique can be both eye opening and inspiring.**

Quick Guide:

- Identify a project and a group of reviewers.
- Pick a time and place for the session.
- Presenter: Describe what has been done and why.
- Reviewers: Ask questions.
- Presenter: Provide clarification.
- Reviewers: Start with warm feedback (positive).
- Reviewers: End with cool feedback (negative).
- Presenter: Invite suggestions from reviewers.
- Presenter: Thank everyone for participating.



Helpful Hints:

- Invite reviews from people who didn't do the work.
- Don't wait for completeness to invite critique.
- Get in the habit of asking for feedback often.

Benefits:

- Facilitates constructive discussion
- Reveals blind spots in your design activities
- Shows opportunities for improvement
- Builds organizational alignment

System Usability

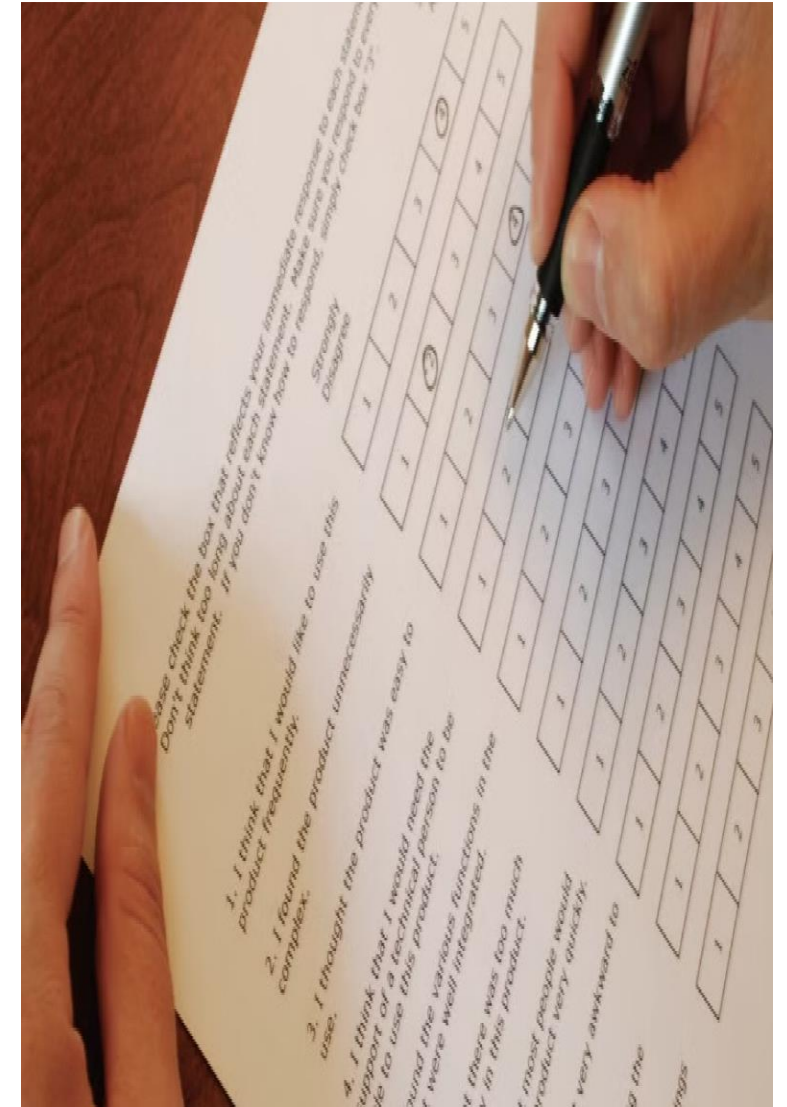
Scale a short survey for quantifying feedback from subjective assessments of usability.

- The way people feel when using something is just as important as how they use it.
- In order to **effectively evaluate a design, then, you may need to measure people's subjective and objective feedback of an experience.**
- To ensure reliability, the **System Usability Scale (SUS) provides a good option.**
- It is a freely available questionnaire originally developed by John Brooke for Digital Equipment Corporation.
- **The SUS uses the Likert Scale, which asks participants to evaluate each question by choosing between five attitude responses, ranging from “Strongly Disagree” to “Strongly Agree.”**

- This is a particularly effective way of benchmarking a given design against later iterations, and is highly versatile across many product and service realms.
- **To score the SUS, first subtract one point from the user response for each odd-numbered item.**
- **Then, for even-numbered items, subtract the user response from five.**
- **This scales all values from zero to four, with a four being the most positive response.**
- **Finally, add up the converted responses and multiply the total by 2.5.**
- **Any score above 68 is considered above average.**

Quick Guide:

- Identify what you will be testing and a few key tasks.
- Invite a group of participants.
- Conduct a task-based usability test.
- Administer the SUS questionnaire after the test.
- Instruct the respondents to answer every question.
- Ask them to record their first response to each item.
- Calculate the total score for each questionnaire.
- Average all of the scores to obtain the overall value.



Helpful Hints:

- Give the participants a printed or an online form.
- Tell them to mark the center point if undecided.
- Don't allow them to think about each item too long.

Benefits:

- Leverages a proven measure of usability
- Standardizes your evaluations
- Provides a manageable numeric score
- Helps you make quick assessments